



Executable Organization

Min Requirement for Executable

1. The `.text` section: Contains the actual executable code
2. The start symbol/address: Usually `__start()`, which calls `main()`

Other Important Sections

- `.data` : Stores global or static variables initialized to non-zero values
- `.rodata` : Contains read-only data
- `.bss` : Reserves space for global/static variables initialized to zero

Programs vs. Processes

- A program is your compiled code (static)
- A process is a program in execution (dynamic)

A process consists of:

1. Instructions and data
2. Current state
3. Resources needed for execution

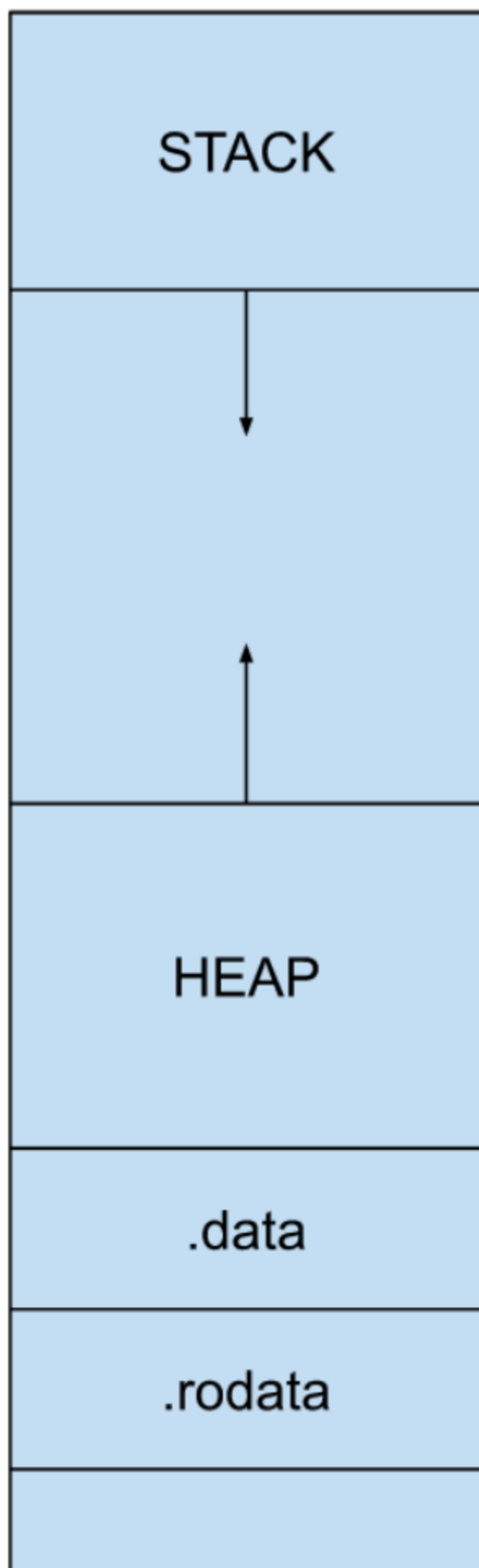
Multiple instances of the same program running simultaneously are separate processes

Memory Assignment and Execution

1. **Compile-time:** The compiler resolves static variables and link
2. **Load-time:** The system sets up memory segments and links shared objects
3. **Run-time:**
 - Dynamic allocation is resolved
 - The code is executed

- The process starts at the start symbol (usually `__start()`)

Executable In Memory



Execution

Once launched the program runs as intended by the programmer, unless the OS needs to intervene.